

AUTHBC

Feasibility Boundaries for Authenticated UAV Telemetry

An Exclusion Bound, a Capacity Envelope, and Hardware Validation

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Abstract

Flying ad-hoc networks (FANETs) stream telemetry records so small that a cryptographic signature (64–96 B) rivals or exceeds the payload itself (45–190 B). Once the payload has been compressed, authentication — not data — becomes the dominant on-air cost, and the operational question stops being *how many bytes does a scheme save* and becomes *which configurations can run at all*.

This thesis answers the second question. It searches four design axes jointly — record *encoding*, authentication *placement*, signature *scheme* and *batch* size — against a byte model, an NS-3-validated IEEE 802.11 broadcast channel model, and energy measured on Raspberry Pi 4 hardware. Three results follow, ordered by how durable they are.

An exclusion. A link whose payload cannot hold one frame header, one signature and one record admits no per-frame-verifiable telemetry at any encoding or batch size, and packet loss forecloses fragmentation as an escape, because a unit spanning n frames verifies with probability $(1 - p)^n$. On the EU868 LoRa band this removes *three* of seven data rates unconditionally: at DR0–DR2 a 64 B signature alone overflows a 51 B payload, so the exclusion survives a zero-byte header and a one-byte record. A fourth rate, DR3, misses by six bytes against a measured frame header of 44 B — and is recovered by an integer-keyed header profile, so the boundary is reported *constructively*: it names not only where authentication cannot fit, but what would have to change.

A capacity envelope. Joining the byte model to the channel model yields the largest neighbourhood each configuration can sustain. The co-designed configuration carries $1.9\times\text{--}3.2\times$ the neighbourhood of an inline-signing baseline, reported across both the saturation and the verifiability threshold rather than at whichever flatters the result.

Bytes, and what they cost. The adopted configuration puts 58.7% fewer bytes on air (72.0 B per record) at an operating point compliant with 3GPP TS 22.125, against a success criterion committed to version control two days before the result existed.

A factorial ablation narrows rather than inflates the co-design claim: only placement and batching interact, by exactly $g_a(1 - 1/b)$; encoding contributes additively and the signature-scheme axis is byte-neutral in this regime. AUTHBC introduces no new cryptographic primitive. Its contribution is a reproducible co-design, its quantified trade-offs, an honest account of the boundary beyond which no choice of existing cryptography helps

— and a methodological record of the defects found while establishing it.

Declaration

► *TO BE COMPLETED by the candidate. Institutional wording required.*

Use of generative AI.

► *The candidate must state this explicitly and accurately; the repository history records the extent of agent involvement in implementation, experiment execution and drafting. Institutional policy governs the required wording.*

Acknowledgements

► *TO BE COMPLETED.*

Nomenclature

Symbols follow docs/01_SYSTEM_MODEL_ARCHITECTURE.md §2, which is the single source of truth for notation across the repository, the papers and this thesis.

Symbol	Meaning	Default
s	encoded record payload bytes	measured (E1)
g_a	authentication object bytes (scheme σ)	64 / 64 / 96
H_f	ledger frame header bytes	38–44, measured
M	application MTU budget	1500 (802.11)
b	records per frame (batch)	decision variable
p	frame loss probability	{0.02, 0.05, 0.10}
ε	verifiability slack; require $V \geq 1 - \varepsilon$	$= p$
Λ_i	per-UAV record rate (rec/s)	50 (adopted)
Λ	aggregate rate a receiver verifies	$\Lambda_i \cdot N_{\text{local}}$
N_{local}	neighbours in the collision domain	reported as a curve
U	channel utilisation, offered / deliverable	admission metric
V	$P(\text{record verifiable at receiver})$	constraint $\geq 1 - \varepsilon$
R	PHY data rate	6 Mbit s ⁻¹
$D(b)$	freshness of the oldest record in a batch	$\leq D_{\text{max}}$
s_{max}	largest record a single frame admits	$M - H_f - g_a$
N_{max}	largest sustainable neighbourhood	derived

A note on H_f . It is reported as a *range*, not a constant. Canonical CBOR encodes small integers in one byte and large ones in five, so the measured frame header moves between 38 and 44 B with the magnitudes of the sender identifier and sequence number. This matters because one of the thesis’s own results depends on it; Chapter 10 states where.

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Chapter 1

Introduction

1.1 The problem, in one paragraph

A flying ad-hoc network of unmanned aerial vehicles must do two things at once that pull against each other. It must broadcast telemetry often enough to be useful — position, velocity, battery, mode, tens of times a second — and it must make that telemetry *tamper-evident*, so that a record cannot be forged, replayed or silently altered by a participant or an outsider. The second requirement costs bytes. A digital signature at the 128-bit security level is 64 B for Ed25519 or ECDSA-P256, and 96 B for the BLS12-381 variant actually shipped by the reference implementation. A telemetry record, once compressed, is 45 B.

That inequality is the whole thesis. **The signature is larger than the thing it signs.**

1.2 Why this is not simply a compression problem

The obvious response — compress harder — fails, and it fails for a reason worth stating precisely. Payload compression was the subject of prior work; this thesis begins where that work ends. Once a record is squeezed from 191 B of JSON to 45 B of delta-encoded CBOR, the authentication object has gone from 25% of the on-air cost to 59%. Every further byte saved on the payload increases the fraction spent on cryptography. Compression does not solve the problem; *compression is what creates it*.

The design space that remains has four axes, and this thesis searches them jointly:

- **Encoding** e — how a record is serialised (JSON, CBOR, MessagePack, delta-CBOR).
- **Placement** — *where* the signature sits relative to the records it covers: inline per

record, self-batched over b of one sender's records, cross-signer aggregated, or block-level across multiple frames.

- **Scheme** σ — which signature algorithm, and therefore how many bytes the authentication object costs and how much CPU verification demands.
- **Batch** b — how many records share one authentication object, bounded by the frame size on one side and by a freshness deadline on the other.

1.3 The question this thesis actually asks

The literature on efficient authentication asks *how many bytes does my scheme save*. That is a useful question and it has been answered many times. This thesis asks a different one, because for the deployments that motivate it the byte count is not the binding constraint:

Which configurations can run at all?

That reframing — from optimisation to feasibility — is the thesis's organising decision, and Chapter 11 records why it was made partway through the work rather than at the start. Briefly: as a co-design optimisation the result is mid-tier, because the optimisation is largely closed-form and a factorial ablation showed only one genuine interaction among the four axes. As a *boundary* result the same evidence is stronger, because an impossibility cannot drift as models improve — and four of this project's own performance numbers did drift, and had to be corrected.

1.4 Research questions

- RQ1** How large is authentication overhead after compression, and where is the payload-size threshold below which it dominates?
- RQ2** Which authentication placement minimises on-air bytes while keeping records verifiable under loss, and what does self-batching change relative to cross-signer aggregation?
- RQ3** When does an aggregatable scheme beat a fast conventional one, as a function of relay fraction, arrival rate and the radio/CPU power ratio?
- RQ4** What is the end-to-end optimal configuration, how much does it beat the naive and compression-only baselines, and do the analytical models survive contention simulation and real hardware?
- RQ5** (*added during the work*) Where is the boundary beyond which no choice of *existing* cryptography is feasible at all — and how much of that boundary is arithmetic rather than an artifact of this design?

RQ5 was not in the original charter. It emerged from the low-rate arm and became the thesis’s most durable result; Chapter 10 develops it and is honest about which part of it survives scrutiny and which part does not.

1.5 Contributions

1. **An overhead-dominance characterisation** for compressed authenticated FANET ledgers, with the threshold stated in closed form and measured across four encodings (Chapter 4, Chapter 7).
2. **A placement theory** distinguishing self-batching from cross-signer aggregation, with a loss-robustness frontier proving that frame-level batching Pareto-dominates block-level under any stationary loss process — not merely under the independent-loss assumption the emulator implements (Chapter 4).
3. **An exclusion bound instantiated on a real band, separated by durability.** Three EU868 data rates are excluded *unconditionally*: a 64 B signature will not fit a 51 B payload at a zero-byte header. A fourth is excluded by *our own* frame header, and the thesis quantifies that contingency rather than leaving it implicit (Chapter 10).
4. **A feasibility envelope** rather than a byte ratio: the largest neighbourhood each configuration sustains, at $1.9\times$ – $3.2\times$ the inline baseline, reported at two thresholds because they answer different questions (Chapter 8).
5. **Validated models with their gaps stated**, including a broadcast channel model validated to $\leq 0.08\%$ against an independent slot-exact simulator and $\leq 0.51\%$ against NS-3, plus two-radio hardware measurement of the airtime constants (Chapter 9).
6. **An ablation that narrows our own claim**, showing that only placement and batching genuinely interact, and that an apparent encoding interaction was a ratio-scale artifact (Chapter 8).
7. **A reproducible testbed and a defect taxonomy.** Every model-derived artifact is re-derived byte-identically by an automated gate, and the classes of defect that survived that gate are catalogued — including the ones found in this work (Chapter 11).

1.6 Explicit non-goals

This thesis introduces no new cryptographic primitive, no new consensus protocol and no post-quantum implementation; post-quantum sizes appear only as analytical projections.

Key compromise, multi-hop routing research and finality semantics are out of scope. The low-rate arm is a *modelling* chapter: no LoRa radio was metered and no energy column exists for it.

1.7 Structure of this thesis

Chapter 2 places the work in its literature. Chapter 3 fixes the system and threat models and the notation. Chapter 4 develops the theory. Chapter 5 describes what was built and Chapter 6 how it was measured. Chapter 7 and Chapter 8 give the 802.11 results; Chapter 9 the channel-model validation and hardware; Chapter 10 the low-rate regime and the exclusion bound. Chapter 11 is unusual for a thesis and deliberate: it reports the errors found in this project's own results, how they were found, and what class of defect resisted the safeguards. Chapter 12 concludes.

Chapter 2

Background and Related Work

► *STATUS: OUTLINED, PARTLY DRAFTED. The positioning below is adapted from the paper’s Related Work, where every reference was verified against the Crossref record for its DOI. A thesis requires materially broader coverage than a 15-page paper: each section below names the reading that must still be done. Do not submit this chapter as it stands. See `thesis/STATUS.md`.*

2.1 Citation integrity in this thesis

One methodological commitment applies to the whole chapter and is stated before any claim is made. Every reference was confirmed against the Crossref record for its DOI rather than from a search summary, and no figure or number is quoted from a source that has not been read in full. Where a source establishes prior art for something this thesis might otherwise have claimed as novel, that is said explicitly.

That rule exists because it was broken once. An early finding claimed this work was the *more optimistic* of two LoRa capacity models; the claim was drawn from a figure in a paper that reported pure-ALOHA results alongside LoRa results, and the wrong figure was read. The finding was retracted and the lesson recorded as a rule: **quoting the paper is not enough — quote the figure**. Chapter 11 returns to this.

2.2 Flying ad-hoc networks and UAV telemetry

► *TO WRITE. Required coverage: FANET as distinct from MANET/VANET (mobility degree, topology change rate, energy asymmetry); the telemetry workload itself, anchored to autopilot source rather than secondary summaries — PX4’s MAVLINK_MODE_ONBOARD at 50 Hz and ArduPilot’s vehicle-specific defaults; and 3GPP TS 22.125 S5.2.2 as the independent standards anchor for direct UAV-to-UAV broadcast. Sources are held; the synthesis is not written.*

2.3 Tamper-evident telemetry and hash-chained ledgers

Nakamoto’s hash chain [1] makes an append-only log tamper-evident: altering a record invalidates every subsequent link. UAV-blockchain systems, surveyed in [2], place telemetry or attestations on such a chain, gaining integrity at a per-record overhead cost. This thesis targets exactly that substrate cost and is agnostic to the consensus layer above it; finality and witnessing semantics are out of scope.

► *TO EXPAND: the distinction between a hash chain (integrity of ordering) and a signature (attribution) is load-bearing for the threat model in Chapter 3 and deserves a full subsection.*

2.4 Signatures for constrained authentication

ECDSA [3] yields 64 B P-256 signatures at 128-bit security but is randomness-sensitive. Ed25519 [4, 5] uses a twisted-Edwards curve with deterministic nonces for fast, misuse-resistant, batchable verification. BLS [6] gives the shortest signatures via pairings, and [7] aggregates n signatures into one at the cost of expensive pairing verification.

AUTHBC introduces *no new primitive*. It quantifies *where and when* each scheme wins once placement and batching are fixed (Chapter 8), finding Ed25519 dominant for self-batching and BLS worthwhile only for cross-signer relay traffic.

Independent work on UAV swarms adopts precisely the design this thesis treats as its naive baseline: Rotta and Mykytyn attach a HMAC-256 signature with a timestamp to *each* broadcast telemetry message, and observe that MAVLink telemetry messages are much smaller than 256 bytes [8] — corroborating both the premise and the baseline from outside this work. Their construction is a symmetric MAC under a group key, so it provides integrity and group authenticity but *not* non-repudiation, and therefore cannot underwrite a ledger attributable to a specific UAV. That is a different trust model, not a cheaper solution to this one.

2.5 Batch and aggregate verification

Zhang *et al.* [9] let a roadside unit verify multiple received signatures simultaneously, motivated by the same pressure from the opposite side: a hard deadline — DSRC’s 300 ms — within which sequential verification does not scale. Their batching is *receiver-side* and buys verification time; the batching in this thesis is *sender-side* and buys airtime. The two are complementary rather than competing. This work generalises from receiver batching to authentication *placement*, co-designs it with encoding and scheme, and adds a loss-robustness frontier that pure aggregation does not address.

► *TO WRITE: certificateless aggregate signature (CLAS) schemes as the closest external comparator. The finding to develop is the axis, not a ratio: published CLAS overheads are linear in*

message count (583–859 B/record) because aggregation compresses the verifier’s work, not the wire. Do not claim to beat CLAS — those schemes buy conditional privacy this work does not offer, and their group element is 128 B against our 64 B. Also outstanding: implicit (ECQV) certificates from IEEE 1609.2, which were never priced here.

2.6 Compact serialisation and delta coding

CBOR [10] and MessagePack [11] give compact binary encodings. Delta coding of telemetry time series is established practice, most prominently in Gorilla [12], which stores delta-of-delta timestamps and XOR-folded values for roughly a 10× reduction.

The variant used here differs in one respect the link forces. Gorilla is a database and can chain deltas indefinitely; a lossy broadcast link makes an unbounded chain fragile, so this work re-anchors on a keyframe every K records and pays the resulting size for loss independence.

2.7 Channel models for IEEE 802.11

Bianchi’s fixed-point analysis of the DCF [13] is the standard model for saturated unicast throughput. Broadcast is *not* that model with the acknowledgement removed — a point Ma and Chen make explicitly, and which this project rediscovered expensively. Their broadcast model [14, 15] accounts for the backoff-counter Consecutive Freeze Process, which dominates when the contention window is small relative to the station count. Tinnirello, Bianchi and Xiao [16] supply the anomalous-slot refinement that explains a residual small-frame bias observed in Chapter 9.

2.8 LoRa and LoRaWAN capacity

Bor *et al.* [17] provide the only published LoRa capacity model found that is grounded in hardware interference measurements rather than an analytical collision assumption, and state it in closed form so it can be evaluated at this work’s operating point rather than merely quoted at theirs. Zirak *et al.* [18] supply the only air-to-air LoRa PDR-versus-range field data located, which range-limits the capacity result of Chapter 10.

► *TO EXPAND: LoRaWAN’s regional parameters as a regulatory rather than a technical constraint — the EU868 duty cycle is enforced per sub-band, which is why adding channels buys no extra airtime. This is developed in Chapter 10 but belongs partly here.*

2.9 Where this thesis sits

The adjacent literature almost uniformly proposes a *new construction* and evaluates it against its predecessors. This thesis deliberately proposes none. Its contribution is a boundary: the set of configurations in which no choice of existing, standardised cryptography is feasible at all. That result is arithmetic rather than empirical for the part that matters most, so unlike a performance figure it does not move as models, hardware or schemes improve — and it is invisible to the byte models the literature optimises against.

► *HONESTY REQUIREMENT for this section. The reference count currently stands at 46, short of the 45–60 a thesis of this scope would normally carry, and it stops there deliberately: 46 is every source held and read. Reaching a higher count requires a genuine sourcing pass — find, download, read, then cite. Padding the list would be the same defect this project spent its audit removing.*

Chapter 3

System Model and Threat Model

► *STATUS: DRAFTED from docs/01. Needs a figure of the frame layout and one worked example of a frame on the wire.*

3.1 Network

N UAVs share an IEEE 802.11 broadcast channel (OFDM, base rate $R = 6 \text{ Mbit s}^{-1}$ for robustness; application MTU budget $M = 1500 \text{ B}$). Frames are lost independently with probability p , an abstraction of fading, collisions and mobility, with $p \in \{0.02, 0.05, 0.10\}$ as a declared sensitivity range. Contention is modelled by the DCF (Chapter 4) and validated against NS-3 (Chapter 9). There is no ground control station in the model: the substrate is peer-to-peer broadcast.

3.1.1 Why broadcast, and why it matters

IEEE 802.11 broadcast frames carry no acknowledgement and are never retransmitted. Three consequences follow, and each shapes a later chapter. A broadcast receiver sees the *raw* channel error rate with none of the ARQ that hides loss on a unicast link, which is why the loss grid is one to two orders of magnitude more pessimistic than the 99.9% reliability TS 22.125 specifies for managed command-and-control links *with* retransmission. Overload degrades *delivery*, never latency, because excess offered load is dropped rather than queued. And the channel model is not Bianchi's — Chapter 9 records that using the unicast model with the acknowledgement removed under-predicts simulation by $16\times$.

Table 3.1: The operating region, with costs stated. Compliance costs swarm size and nothing in bytes.

	Λ_i	$D_{\max}$	b	B/rec	$N_{\max} (U < 1)$	$N_{\max} (V \geq 0.95)$	TS 22.125
adopted	50	100 ms	4	72.0	35	100	yes
alternative	20	250 ms	4	72.0	103	213	no
—	20	100 ms	1	153.0	34	98	yes, no batching
—	10	100 ms	1	153.0	78	180	yes, no batching

3.2 Ledger

Each UAV maintains an append-only hash chain of its own telemetry records and stores verified records overheard from others. A record is

$$\text{rec} = \{\text{src}, \text{seq}, \text{ts}, \text{prev_hash}, \text{payload}\}, \quad \text{prev_hash} = \text{SHA-256}(\text{previous record bytes}).$$

Only the owner appends to its own chain — an owned-object model — so tampering or equivocation on a (src, seq) pair is detectable by chain verification. Finality and witnessing semantics are out of scope; the deliverable is *authenticated dissemination*, not consensus.

3.3 Traffic and the two arrival rates

Records are generated at rate Λ_i per UAV. The aggregate neighbourhood arrival rate is $\Lambda = \Lambda_i \cdot N_{\text{local}}$ — the rate a *receiver* must verify.

Remark. These are different quantities and conflating them is a real defect, recorded as such. A node batches its *own* records, so freshness uses Λ_i ; it verifies *everyone's*, so the verification-throughput constraint uses Λ . Using Λ_i for both silently models a one-sender network and makes cross-signer aggregation look feasible at fleet sizes where the receiver cannot keep up.

3.3.1 The adopted operating point

$\Lambda_i = 50 \text{ rec/s}$ at $D_{\max} = 100 \text{ ms}$ is adopted. It is PX4's MAVLINK_MODE_ONBOARD companion-computer rate, read from autopilot source rather than from a secondary summary, and it is fully compliant with 3GPP TS 22.125 S5.2.2, which specifies at least 10 messages per second and at most 100 ms end-to-end latency for direct UAV-to-UAV local broadcast.

The alternative (20 rec/s, 250 ms) point is retained and reported because it gives *identical bytes* — batching obeys $b \leq \Lambda_i D_{\max}$, so only the product matters, and both give $\Lambda_i D_{\max} = 5$, hence $b = 4$ and 72.0 B per record.

3.3.2 The floor of the region, which is a real constraint on the application

Holding $b \geq 4$ under a 100 ms deadline requires $\Lambda_i \geq 40$ Hz. At 20 or 10 Hz the same deadline forces $b = 1$ and the saving collapses from 58.7% to 12.2%. **The batching benefit is not available at every compliant operating point:** it requires a telemetry rate fast enough to fill a batch inside the deadline. More structurally, any batching at all needs $\Lambda_i \geq 1/D_{\max}$ — at the standard’s 100 ms bound, exactly the standard’s own minimum message rate. At an ArduPilot Copter default of 0 Hz, where streams are requested by a ground station rather than pushed, this thesis’s mechanism does not apply at all.

3.4 Threat model

The adversary may inject, replay, modify or forge frames, but does not hold any honest UAV’s private key; key compromise is out of scope by charter. The security requirements are existential unforgeability under chosen-message attack at the 128-bit level, replay rejection via monotonic (`src, seq`) per chain, and integrity via the hash chain.

Remark. Availability under loss is a *robustness* metric in this thesis, not a security claim. The quantity V , the probability that a record is verifiable at a receiver, is treated throughout as a design constraint rather than as a guarantee against an adversary who controls the channel.

3.5 The decision space

3.5.1 Authentication placements

- A — inline per record.** Every record carries its own signature. The baseline.
- B — self-batch.** One UAV packs b of its *own* records into a frame and signs the frame once. Per-record authentication cost $(g_a + H_f)/b$. Self-contained, hence loss-local.
- C — cross-signer aggregate.** A frame carries b records from *different* originators, each originally signed; the signatures aggregate into one. Required only when forwarding others’ records or combining attestations.
- D — block-level aggregate.** One signature over b records spanning $n > 1$ frames. All-or-nothing under loss; Chapter 4 shows when this is dominated.

3.5.2 Schemes and encodings

Signature schemes are ECDSA-P256 (64 B, legacy baseline), Ed25519 (64 B, fast) and BLS12-381. The BLS object is 96 B, not the 48 B minimum-signature variant originally specified: the available library ships only the minimum-pubkey scheme. That deviation

Table 3.2: H_f is not a constant. Canonical CBOR encodes integers 0–23 in one byte and those ≥ 65536 in five, so the measured header moves with field magnitude.

src	base_seq	H_f
0	0	38 B — first records of a flight, low node id
24	24	39 B
256	256	40 B
40 000	180 000	44 B — one hour into flight at 50 Hz

is recorded and applied consistently, and it makes BLS lose the byte comparison outside multi-signer aggregation.

Encodings are JSON (baseline), canonical CBOR, MessagePack, and delta-CBOR with keyframes every $K = 16$ records. K is fixed and not optimised — a real design variable left on the table, and declared as such.

3.6 The frame header, which is a range

H_f feeds the batching theory, the exclusion bound, the batch bounds, the channel-utilisation constraint and the energy model. It is measured directly from the implemented wire format:

$$H_f = \text{len}(\text{encode_frame}(F)) - \sum \text{len}(\text{record canonical bytes}) - \text{len}(\text{auth}).$$

The value 44 B is used throughout, and it is the steady-state value for any realistic flight. It is also the *top* of the measured range, and Chapter 10 shows that this matters: the exclusion bound is $s_{\max} = M - H_f - g_a$, so a larger header makes exclusion *more* likely. Using the top of a measured range is the choice most favourable to that result, and the thesis says so where the result is stated.

Remark (Known headroom, not claimed). The empty frame skeleton is 43 B, of which 29 B are CBOR *text key names*. An integer-keyed profile costs 7 B for the same seven keys, giving $H_f = 22$ B. Additionally, every record carries **src** and **seq** while the frame already carries **src** and **base_seq**; in placement B all records share a sender and have consecutive sequence numbers, so both fields are derivable — a further 10 B per record. Neither optimisation is adopted, because the wire format is frozen for reproducibility. Both are *measured* rather than merely admitted, which makes every byte figure in this thesis an upper bound on an untuned design. Chapter 10 shows that one of the thesis’s own results turns on this.

Chapter 4

Theoretical Framework

► *STATUS: DRAFTED from docs/02. Proofs are complete for T1–T3 and T6; T5 is stated honestly as an empirical separability result rather than a theorem, which is what it is.*

This chapter develops the closed-form results. Each carries its empirical validation hook, and where a result is *applied* rather than novel, that is stated at the point of statement rather than buried in related work.

4.1 T1 — the overhead-dominance threshold

Theorem 4.1 (Overhead dominance). *With inline per-record authentication, the authentication byte fraction is $\varphi(s, g_a) = g_a/(s + g_a)$, and $\varphi \geq \alpha \iff s \leq g_a(1 - \alpha)/\alpha$. In particular authentication dominates ($\alpha = \frac{1}{2}$) exactly when $s \leq g_a$.*

Proof. $\varphi \geq \alpha \iff g_a \geq \alpha(s + g_a) \iff g_a(1 - \alpha) \geq \alpha s$. □

The threshold is trivial; its *instantiation* is the motivating result. At $g_a = 64$ B and measured record sizes, JSON at 191 B gives $\varphi = 25.1\%$, CBOR at 66 B gives 49.1%, and delta-CBOR at 45 B gives 58.7%. Compression moved the bottleneck from data to cryptography. Validated by E1 (Chapter 7).

4.2 T2 — frame-level batching and the amplification law

Placement B or C packs b records, one authentication object of g_a bytes and a header of H_f bytes into one frame, subject to $b \leq b_{\max}(s) = \lfloor (M - H_f - g_a)/s \rfloor$.

Proposition 4.2. *Per-record authentication overhead $\omega(b) = (g_a + H_f)/b$ is strictly decreasing, so the byte-minimal feasible batch is $b = b_{\max}(s)$. Relaxing integrality, per-record on-air bytes are $C(s) = s \cdot A$ with amplification $A = M/(M - H_f - g_a) \geq 1$.*

Proof. $\omega'(b) < 0$ gives the first claim. For the second, substitute $b = (M - H_f - g_a)/s$ into $s + \omega(b)$ and factor. $\square$

So every payload byte saved by compression saves A on-air bytes — *in the MTU-limited regime*. This is MTU-efficiency algebra, not a new law, and Theorem 4.3 shows the regime is never reached on 802.11.

4.3 T2a — which ceiling binds, and what compression is therefore worth

Theorem 4.3 (Binding constraint). *Let $b^* = \min(b_{\text{MTU}}, b_{\text{fresh}})$ with $b_{\text{MTU}} = \lfloor (M - H_f - g_a)/s \rfloor$ and $b_{\text{fresh}} = \lfloor \Lambda_i D_{\text{max}} \rfloor$. Then*

$$\frac{dC}{ds} = \begin{cases} A = \frac{M}{M - H_f - g_a}, & \text{MTU-limited,} \\ 1 \text{ exactly,} & \text{freshness-limited.} \end{cases}$$

Freshness binds exactly when $s < (M - H_f - g_a)/(\lfloor \Lambda_i D_{\text{max}} \rfloor + 1)$.

Proof. Differentiate C in each regime; in the second, b is constant in s . $\square$

The consequence is sharp and negative for the 802.11 arm. At $M = 1500$, $H_f = 44$, $g_a = 64$, $\Lambda_i D_{\text{max}} = 5$, the boundary is $s < 232.7$ B. Every encoding studied lies below it — delta 45, MessagePack 65, CBOR 66, JSON 191 — so **freshness binds for all of them and A is never operative on 802.11**. Compression pays exactly $1\times$, and the residual authentication cost is a floor $(g_a + H_f)/(\Lambda_i D_{\text{max}})$ that compression cannot touch.

On a low-rate link ($M = 242$) the boundary falls to $s < 29.7$ B, so the MTU binds and A is operative. **The leverage that motivates compression is exclusive to the low-rate arm**, which strengthens rather than weakens that chapter's motivation.

4.4 T3 — the loss-robustness frontier

Frame-level placements are self-verifiable, so $V_B = 1 - p$ and a loss costs b records. Block-level placement D spans $n(b) = \lceil (bs + g_a + H_f)/M \rceil$ frames under one signature, so $V_D = (1 - p)^n$.

Theorem 4.4 (Fragmentation is foreclosed). *The constraint $V \geq 1 - \varepsilon$ admits at most $n_{\text{max}} = \lfloor \ln(1 - \varepsilon)/\ln(1 - p) \rfloor$ frames. If $\varepsilon \leq p$ then $n_{\text{max}} = 1$: block-level authentication is infeasible beyond a single frame.*

Proof. Solve $(1 - p)^n \geq 1 - \varepsilon$ for n . At $\varepsilon = p$ the right-hand side is $1 - p$ and $n \leq 1$. $\square$

Theorem 4.5 (Robustness to correlated loss). *For any stationary loss process of mean rate p , whatever its correlation structure,*

$$V_D(n) = P(\text{all } n \text{ frames arrive}) \leq P(\text{one given frame arrives}) = 1 - p = V_B,$$

with equality only at $n = 1$.

Proof. A joint probability cannot exceed a marginal. □

This matters more than it first appears. The $(1 - p)^n$ form assumes independent loss, which the emulator also implements — so agreement between measured and theoretical V is a *consistency check, not a validation*. Theorem 4.5 rescues the ordering without the assumption: independence is the *worst case* for D, and correlation narrows the gap but cannot invert it. Gilbert–Elliott simulation at matched mean $p = 0.05$ confirms it: $V_D(n=2)$ rises from 0.9025 (independent) to 0.9498 at a mean burst of 160 frames — asymptotically $1 - p$, never past it, and never reaching the feasibility threshold.

4.5 T4 — scheme selection

For a UAV’s *own* records under self-batching, one ordinary signature covers b records, so Ed25519 and BLS are near-equivalent in bytes while Ed25519’s verification is far cheaper than a pairing. Ed25519 wins self-batching outright.

For *relayed* records from distinct signers, Ed25519 must carry $b \cdot 64$ B while BLS aggregates. BLS saves $\Delta(b)$ bytes per record but costs Δt extra CPU. BLS is energy-optimal iff

$$P_r \cdot \frac{8\Delta(b)}{R} > P_c \cdot \Delta t,$$

and throughput-feasible iff $t_{\text{agg-verify}}(b)/b \leq 1/\Lambda$. At $R = 6 \text{ Mbit s}^{-1}$ the radio side is $\approx P_r \cdot 75 \mu\text{s}$ against a CPU side of $P_c \cdot (1\text{--}3) \text{ ms}$, so **Ed25519 wins on 802.11 and BLS’s regime is low-rate links and attestation compression** — a falsifiable, regime-dependent rule.

4.6 T5 — the co-design result, stated as what it is

Remark (Not a theorem). T5 is often written as a co-design theorem. It is not one. The joint optimum is *empirically separable*, verified by exhaustive search over the full grid of 4 encodings $\times$ 3 schemes $\times$ 4 placements $\times$ 32 batches = 1536 points. A non-separable optimum would have been found had one existed; none was. That is evidence, not proof, and the distinction is kept.

The recipe: pick the smallest deterministic encoding e^* , placement B for own traffic

or C for relay, batch $b^* = \min(b_{\max}(s), \lfloor \Lambda_i D_{\max} \rfloor)$ — on 802.11 the second term binds — and scheme per T4.

A factorial ablation (Chapter 8) narrows this further, and against the project’s own earlier claim: only placement and batching genuinely interact, by exactly $g_a(1 - 1/b)$, hence not at all at $b = 1$. Encoding is perfectly separable and the scheme axis is byte-degenerate here.

4.7 T6 — the authentication-exclusion threshold

Remark (Prior art, checked before publication). The inequality below is *established*, not novel. Gündoğan *et al.* compute exactly $M - H_f - g_a = s_{\max}$ for 802.15.4/NDN, and the post-quantum literature reports NIST signatures as incompatible with 5G SIB1’s 372 B limit. That a fragmented unit delivers with $(1 - p)^n$ is likewise standard in 6LoWPAN. T6 is therefore a *synthesis* of two known facts, presented as an applied threshold. What has not been found stated elsewhere is the *composition*: that the sliced-signature workaround the literature proposes is foreclosed whenever $\varepsilon \leq p$. This check was run *before* publication precisely because an earlier novelty claim in this project was not.

Theorem 4.6 (Exclusion). *A frame that is independently verifiable must carry the header, the authentication object and at least one record. A link of maximum payload M therefore admits authenticated telemetry iff*

$$s_{\max}(M) = M - H_f - g_a \geq s_{\min},$$

and no choice of batch size, chain placement or aggregation changes this: those decide only how the space above the floor is used.

Fragmentation does not escape it, by Theorem 4.4: at $\varepsilon \leq p$ the entire verifiability budget is spent on the first frame, so fragmenting is already a constraint violation and T6 is evaluated at $n = 1$.

4.7.1 Three tiers, because each names a different escape route

tier	condition	what could fix it
signature	$M < g_a$	a smaller signature scheme <i>only</i>
framing	$g_a \leq M < H_f + g_a$	a leaner frame header
encoding	$H_f + g_a \leq M < H_f + g_a + s_{\min}$	a smaller record encoding

The tier structure is not decoration: it is what makes the bound *constructive*, and Chapter 10 shows the encoding tier being attacked successfully — at the cost of one of this thesis’s own claims.

4.8 A withdrawn theorem, kept visible

An earlier version of this framework contained a seventh result, T7, claiming that channel capacity excludes configurations at $U \geq 1$, where U is offered load over *saturation* throughput. **It was withdrawn on the day it was written**, by its own validation experiment.

Saturation throughput is what the channel carries when every node is backlogged; a lightly loaded channel collides far less and carries substantially more. Simulation at $N = 50$ shows 98.8% of frames still delivered at $U = 1.00$, with the verifiability crossing near $U \approx 2.4$. The claimed exclusion does not exist, and a “50% ceiling” derived from it was an artifact of the wrong threshold.

A second structural error compounded the first: the theorem assumed overload manifests as delay. It does not. Broadcast has no ARQ and no queue buildup, so excess load is dropped, not queued — mean delay at nine times saturation is still under 3 ms. **Any capacity argument here must be made in V , not in D .**

What survives is a cost rather than an exclusion: because $b \leq \lfloor \Lambda_i D_{\max} \rfloor$, tightening the freshness deadline shrinks the batch and raises the frame rate the medium must carry. Meeting the standard’s deadline costs roughly a factor of two in supportable swarm size and *nothing* in bytes. That is the honest, measured statement, and it is a better result than the theorem it replaces.

Remark (The lesson, recorded). T7 was written from a model quantity treated as a physical limit, and published into the documentation before the experiment that tested it existed — an experiment already scheduled. Running it first would have prevented the claim.

4.9 Freshness, and a convention that had to be derived

The freshness of the oldest record in a batch is modelled as $D(b) = b/\Lambda_i + T_{\text{air}} + W_q$ and enforced as a hard constraint.

Remark (Which quantity b/Λ_i is). With periodic arrivals, b/Λ_i is the *batch-window duration* — open to transmit — while the *age of the oldest record* at transmission is $(b-1)/\Lambda_i$, because it waits out the remaining $b-1$ gaps. Both were confirmed exactly by discrete-event simulation under deterministic and Poisson arrivals. The formulation above names the second and computes the first.

It is retained as the *worst case*, and it is exactly that: a record timestamped at its sample instant may describe state up to $1/\Lambda_i$ older, and $(b-1)/\Lambda_i + 1/\Lambda_i = b/\Lambda_i$. So it bounds end-to-end latency including sampling quantisation. The cost of the choice is stated rather than hidden: the tight reading admits $b = 5$ at both operating points (66.6 B/record, -61.8%) where the worst case admits $b = 4$ (72.0 B, -58.7%). **This thesis under-reports its own saving by about three percentage points.**

Chapter 5

Implementation

► *STATUS: DRAFTED from docs/05. Needs a module-dependency figure.*

5.1 Design rule: one home per fact

The implementation follows one architectural rule that is worth stating because it is what makes the reproducibility claims of Chapter 11 possible. **Every quantity has exactly one definition, and everything else refers to it.** There is one function that produces the canonical bytes a signature covers; one airtime implementation shared by the channel model, the energy model and the simulation comparison; one place where experiment parameters live.

Where that rule was broken, defects followed. A signature-size constant duplicated between two modules produced a stale result that survived a re-freeze. An airtime approximation that existed in two forms produced a 12% discrepancy on acknowledgement frames.

5.2 The library

encodings/ The e axis: JSON, canonical CBOR (RFC 8949 S4.2), MessagePack, and delta-CBOR with keyframes every $K = 16$. Delta encoders are *stateful*; a fresh instance per record emits all keyframes and silently inflates sizes — a trap that cost real time.

crypto/ The σ axis: Ed25519 and ECDSA-P256 (both 64 B), BLS12-381 (96 B). Known-answer tests from RFC 8032, NIST CAVP and the IETF BLS draft are vendored and must pass before any timing work.

ledger/ The data structure being authenticated: the canonical record, the hash chain, persistence, and verification with replay and equivocation detection.

placement/ The placement axis. `wire.py` is the frame format and the one place that decides *what bytes a signature covers* — the audit boundary. `framer.py` holds the header constant and the batch bounds. `wire_profile.py` measures an integer-keyed *alternative* to the frozen format; it is a measurement and never the shipped format, and a test fails if it leaks into one.

models/ The analytical layer: the DCF fixed point with exact OFDM-symbol airtime, the published broadcast model, the energy and freshness models, the Pareto search with the exclusion tiers, the scheme crossover, and the low-rate PHY with every constant transcribed from primary specifications.

sim/ A slot-exact backoff simulator written *before* the published broadcast model was located, retained as a cross-check that shares none of that model’s assumptions. It earns its place in Chapter 9.

5.3 Wire format

$$\text{Frame} := \{v, t, \text{src}, \text{base_seq}, n, \text{recs}[], \text{auth}\}$$

in canonical CBOR, with the authentication block shaped by placement. The signature input is the canonical bytes of the covered region, and test vectors are frozen.

► *TO ADD: a byte-level worked example of one frame, which would make Chapter 10’s header argument immediately legible.*

5.4 The two-layer data pipeline

The pipeline splits deliberately. **Measured** artifacts — hardware timings, simulation outputs — are non-deterministic, so they are measured once under controlled conditions, committed with environment and configuration-hash headers, and never silently re-rolled. **Derived** artifacts are byte-stable functions of those, so an automated gate re-computes every one of them on every run and fails if a single row differs.

This buys reproducibility at the price of *silent staleness* risk: a frozen artifact can go stale when an upstream decision changes and the artifact is not re-run. That is not hypothetical — it is exactly what produced the project’s first recorded defect. The gate exists because the risk is real.

Chapter 6

Experimental Methodology

► *STATUS: DRAFTED from docs/04. The statistical rules below are binding on every result in Chapter 7–Chapter 10.*

6.1 Binding statistical rules

Every experiment obeys the following, and where a result predates a rule that is recorded rather than retrofitted:

- At least 30 seeded repetitions per configuration, with seeds logged in the artifact.
- Report median and bootstrap 95% confidence interval over 10^4 resamples.
- Microbenchmarks: $\geq 10^4$ iterations after 10^3 warm-up, garbage collection disabled, runs spanning a suspend discarded.
- Comparisons state effect size; never bare means.
- Raw artifacts are immutable once frozen.
- Any model-versus-measurement deviation above 10% must be investigated and either explained in writing or the model corrected. **Silent tolerance widening is forbidden.**

6.1.1 Why 30 seeds, and what it does not buy

The 30-seed rule is not arbitrary and is not original policy — it was imposed *after* four headline numbers were found to have been distorted by small-sample means evaluated against thresholds. None was a modelling error; every one was sampling.

Remark (The limit of the rule). More seeds fixes exactly one class of defect: a small-sample mean landing on the wrong side of a threshold. It does nothing for a threshold applied to

a mean rather than a distribution, for an unverified constant on the measurement path, for a configuration change that perturbs the random realisation, or for a claim wider than the experiment supports. Chapter 11 catalogues instances of each, several found in this project’s own work *after* the 30-seed rule was in force.

6.1.2 Thresholds must be applied to distributions

A criterion of the form $V \geq 0.95$ is a *reliability* requirement. Applying it to a mean across seeds answers a throughput question instead. Where this thesis reports a threshold crossing it reports the distribution alongside — minimum, maximum, standard deviation, and the count of seeds that individually fail — and where the mean and per-realisation readings disagree, both are given.

6.2 Pre-registration

Where a run tests a prediction, the prediction is committed to version control *before the data exist*, as its own commit, so the ordering is third-party checkable in the repository history.

This practice is enforced rather than aspirational, and it has cost the project a claim: one pre-registration assertion was *withdrawn* rather than reconstructed, because writing an expectations file after results are known manufactures evidence. It has also exposed a subtler defect in this author’s own practice, recorded in Chapter 9: a pre-registered *directional* prediction that the experiment had no statistical power to resolve.

► *TO ADD: a table of every pre-registration in the project with its commit hash, the prediction, and the outcome. The material exists in docs/; the table does not.*

6.3 Experiment matrix

Exp	Question	Primary output
E1	overhead dominance (T1)	bytes/record, φ per encoding
E2	batching cure (T2/T2a)	φ and bytes/rec vs b ; binding-ceiling check
E3	loss frontier (T3)	measured V vs $(1 - p)^n$; goodput; Pareto plot
E4	scheme selection (T4)	energy/record, verification wall, crossover
E5	co-design end-to-end (T5)	bytes/energy/goodput vs baselines
E6–E7	low-rate arm and its co-design	feasibility partition, Pareto front
E8	capacity envelope	channel utilisation, $N_{\max}$

Baselines throughout are A+JSON (naive), A+CBOR (compression-only) and D over-

aggregated.

6.4 Simulation campaign

The simulation scenario uses IEEE 802.11a ad-hoc at 6 Mbit s^{-1} , RTS/CTS off, with N saturated senders in *both* unicast and broadcast modes, frame sizes taken from the real framer output. Transport is a packet socket rather than a flow monitor, because MAC-level goodput with no addressing artifacts is what the models predict.

Remark (The comparison rule). Each mode is compared *only* to its matching analytic variant: unicast to the acknowledged DCF model, broadcast to the published broadcast model. Mixing them is the single most effective way to fabricate a model gap, and this project did exactly that for several phases before catching it.

6.5 Hardware campaign

Two Raspberry Pi 4 units, headless with the performance governor pinned. Timings are re-run from the microbenchmark suite; energy uses external current sensors on a separate measurement host with a GPIO synchronisation line, so the benchmarked device is not instrumented by itself.

The energy harness prints its *predicted* value before reading the meter, deliberately, so a measurement cannot be quietly fitted to expectation.

► *TO ADD: the measurement uncertainty budget for the energy rig. The residual model gap is stated in Chapter 9 but its error bars are not derived.*

Chapter 7

Results I — Bytes, Placement and Loss

► *STATUS: DRAFTED. Every number here is reproduced from a committed artifact by an automated gate; figures are regenerated from frozen data.*

7.1 E1 — authentication dominates after compression

Thirty seeds of one thousand seeded telemetry records per encoding, inline placement, $g_a = 64$ B.

Table 7.1: Measured record size and authentication fraction. Compression moves the bottleneck.

encoding	mean bytes	φ at $g_a=64$	role
JSON	191.09	25.1%	naïve baseline
CBOR	66.25	49.1%	compression-only baseline
MessagePack	65.16	49.6%	—
delta-CBOR	45.00	58.7%	the co-design optimum

This is Theorem 4.1 instantiated: at delta encoding the signature is *larger* than the record it signs. The motivating claim of the thesis is a measurement, not a conjecture.

Remark (s is not a property of the encoding alone). The sizes above are measured over the first ~ 50 seconds of each simulated flight. Sequence numbers and timestamps grow with the record index, and variable-length encodings charge for the extra digits, so JSON, CBOR and MessagePack all grow by 2–4 B over a ten-times-longer window while delta-CBOR stays flat — it encodes differences. The bootstrap interval quoted for CBOR is ± 0.02 B and describes *seed* variation; the systematic dependence on window length is ≈ 2.7 B, roughly $150\times$ wider. The direction is conservative: a longer flight inflates the baseline and leaves the optimum alone, so the reported saving is the pessimistic end.

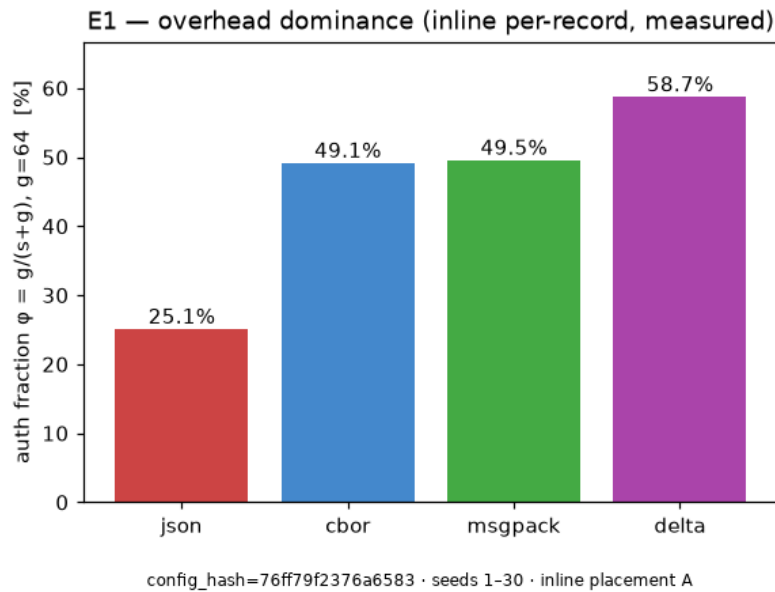


Figure 7.1: Authentication fraction across encodings. Every byte saved on the payload raises the fraction spent on cryptography.

7.2 E2 — batching is the cure, and which ceiling binds

Sweeping placement A against B across encodings and MTU budgets confirms Proposition 4.2 qualitatively and Theorem 4.3 quantitatively. At $M = 256$ the MTU binds and amplification is operative; at $M = 576$ and $M = 1500$ freshness binds and the marginal value of compression is 1.0000 to twelve decimal places.

The practical consequence. On 802.11 the amplification law that motivates aggressive compression is never operative. Compression still pays — it pays exactly $1\times$.

7.3 E3 — the loss frontier

Measured verifiability against batch size and loss probability, placement B versus D. Frame-level batching Pareto-dominates block-level for all $V > (1 - p)^2$, and at $\varepsilon = p$ block-level is infeasible beyond one frame, as Theorem 4.4 requires.

Remark (What this experiment does and does not validate). The emulator implements independent per-frame loss, which is also what the closed form assumes. Agreement is therefore a *consistency check*, not a validation of the loss model. What rescues the ordering is Theorem 4.5, which holds for any stationary loss process, plus Gilbert–Elliott simulation at matched mean rate showing V_D approaching but never reaching $1 - p$.

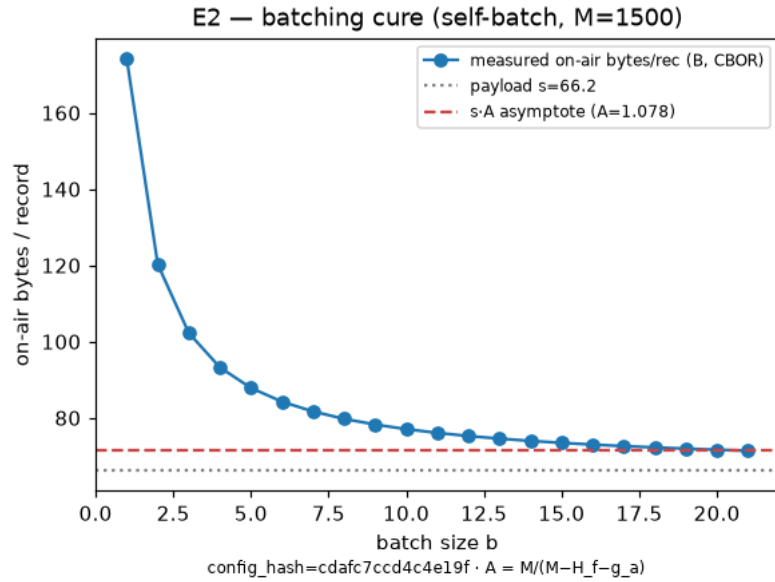


Figure 7.2: Per-record cost against batch size. The curve flattens where the freshness ceiling takes over from the frame-size ceiling.

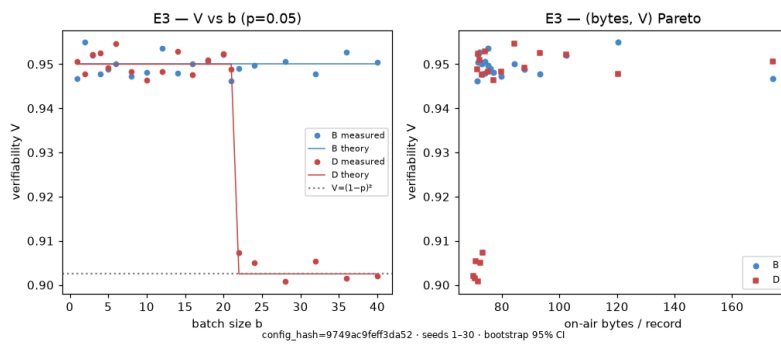


Figure 7.3: The loss-robustness frontier. Block-level authentication trades verifiability for bytes at a rate the constraint does not permit.

Chapter 8

Results II — Co-design and the Capacity Envelope

► *STATUS: DRAFTED. The ablation section deliberately narrows the thesis’s own claim.*

8.1 E5 — the end-to-end configuration

Exhaustive search over the full grid returns delta-CBOR, Ed25519, placement B, $b = 4$: 72.0 B per record against 174.3 B for the compression-only baseline and 299.1 B for the naive one — a 58.7% **total-byte reduction**.

Remark (Report the decomposition, never the bare authentication ratio). The authentication-byte cut is 75.0%, and it is tempting to lead with it. It must not be led with, because it reduces algebraically to $1 - 1/b$: invariant to the header, the signature size, the encoding and the scheme. It is the definition of batching, not a result. The honest headline is the *total* byte reduction reported as a decomposition — placement×batching 79.2%, encoding 20.8%, scheme byte-neutral.

8.2 The factorial ablation, which narrows the claim

A full factorial ablation over the four axes asks whether “co-design” is doing work that single-axis optimisation would not.

- **Placement and batching couple exactly**, by $g_a(1 - 1/b)$ — hence *exactly zero* at $b = 1$, where placements A and B are byte-identical on a single-record frame.
- **Encoding is perfectly separable**. Every interaction term is exactly zero, because s enters additively and therefore *cannot* interact.

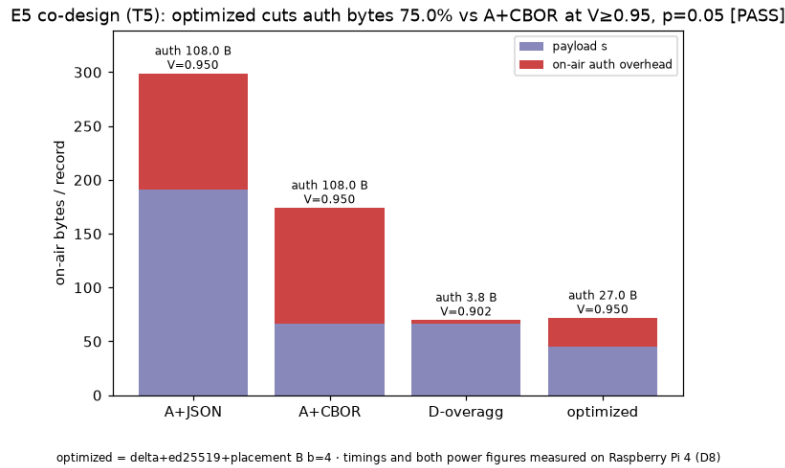


Figure 8.1: End-to-end per-record cost against the two baselines.

- **The scheme axis is byte-degenerate** in this regime: the two 64 B schemes differ only in CPU cost.

An earlier claim that a smaller payload increases the value of batching was a **ratio-scale artifact**: the absolute saving is 81.0 B for every encoding. The four-axis co-design claim was overstated and is now precise. **No number moved — only its interpretation.**

Remark. Note that $1 - 1/b$ is the same term as the warning above. The genuine coupling and the flattering headline are the same algebra, seen from two directions.

8.3 E4 — scheme selection on real hardware

On ARM — the deployment platform — Ed25519 verification is $33.2\times$ cheaper than a BLS pairing, which keeps Ed25519 optimal for self-batching at any plausible radio/CPU power ratio. The ordering between Ed25519 and ECDSA-P256 *flips* between x86 and ARM, so the choice is platform-specific; the authentication-byte result is identical either way, since both are 64 B.

8.4 E8 — the capacity envelope

Joining the byte model to the validated broadcast channel model turns “how many bytes?” into the question an operator actually asks: *how many UAVs, at what telemetry rate?*

The four ratios are $1.94\times$, $3.23\times$, $3.22\times$ and $2.42\times$.

Remark (Do not round this to “about three times”). The spread is $1.9\text{--}3.2\times$ and must be quoted as a range. An earlier draft of this work said “ $\approx 3\times$ holds across readings”, which is both false and precisely the kind of claim the rest of this thesis exists to avoid.

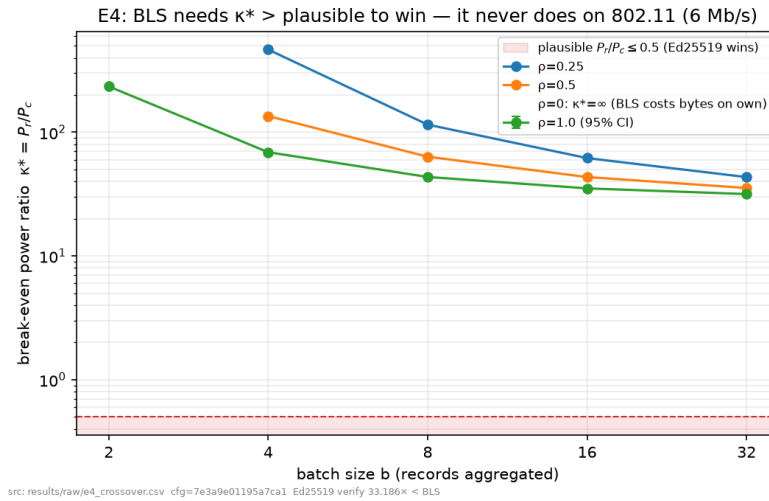


Figure 8.2: The scheme crossover. BLS becomes worthwhile only in the cross-signer regime.

Table 8.1: Largest sustainable neighbourhood, reported at two thresholds because they answer different questions. The per-run column is the reliability reading of the same criterion.

configuration	Λ_i	$D_{\max}$	b	$N_{\max}$ ($U < 1$)	$N_{\max}$ (V mean)	(V per run)
<i>Primary: compliant with 3GPP TS 22.125</i>						
optimized delta/B	50	100 ms	4	35	100	64–91
A+CBOR (compression-only)	50	100 ms	1	18	31	24–29
<i>Relaxed freshness: same bytes, larger swarm, exceeds the standard</i>						
optimized delta/B	20	250 ms	4	103	213	163–201
A+CBOR	20	250 ms	1	32	88	54–80

The co-design advantage is *insensitive to the choice of threshold* — that is the defensible statement — but it is not a single number.

8.5 What the envelope rests on

The absolute capacities depend on a utilisation ceiling measured at one operating point and applied across the envelope. Chapter 9 reports the experiment that tested that assumption, and what it still leaves untested. The *ratios* are protected by construction, because numerator and denominator use the same ceiling; the absolute figures are not, which is why both are given.

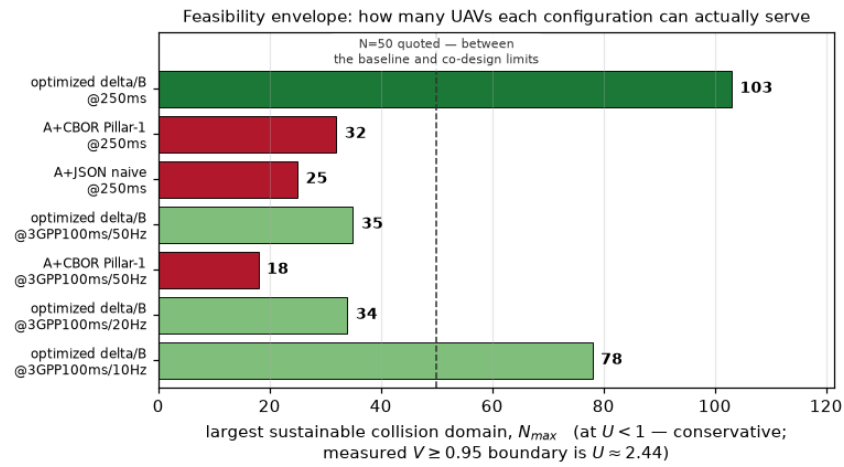


Figure 8.3: The feasibility envelope. The compression-only baseline runs out of channel where the co-designed configuration does not.

Chapter 9

Model Validation and Hardware

► *STATUS: DRAFTED.*

9.1 Three independent implementations of the same physics

The broadcast channel model is the load-bearing input to the capacity envelope, so it is validated against two independent sources rather than one.

Table 9.1: Saturation throughput (Mb/s), 1400 B broadcast frames. The slot-exact simulator was written *before* the published model was located and shares none of its assumptions.

N	published model	slot-exact sim	Δ	NS-3	Δ
5	4.3329	4.3305	−0.06%	4.3318	−0.03%
10	3.1319	3.1321	+0.00%	3.1288	−0.10%
20	1.7823	1.7830	+0.04%	1.7866	+0.24%
35	1.2407	1.2407	−0.00%	1.2343	−0.51%
50	1.2595	1.2585	−0.08%	1.2541	−0.43%

Agreement is $\leq 0.08\%$ against the independent simulator and $\leq 0.51\%$ against NS-3. Unicast agrees with the acknowledged DCF model to $-0.40/+1.29\%$ across $N = 5\text{--}50$, with fixed-point residuals below 10^{-12} .

9.2 The mechanism, confirmed by removing it

The reason broadcast needs its own model is a specific asymmetry. A station that has just transmitted redraws its backoff and may draw zero, taking the medium immediately after the interframe space, while every station that deferred necessarily holds a counter of at least one. In unicast the acknowledgement timeout blocks colliders from doing this; in broadcast nothing does.

The slot-exact simulator can disable *exactly* that asymmetry and nothing else. Doing

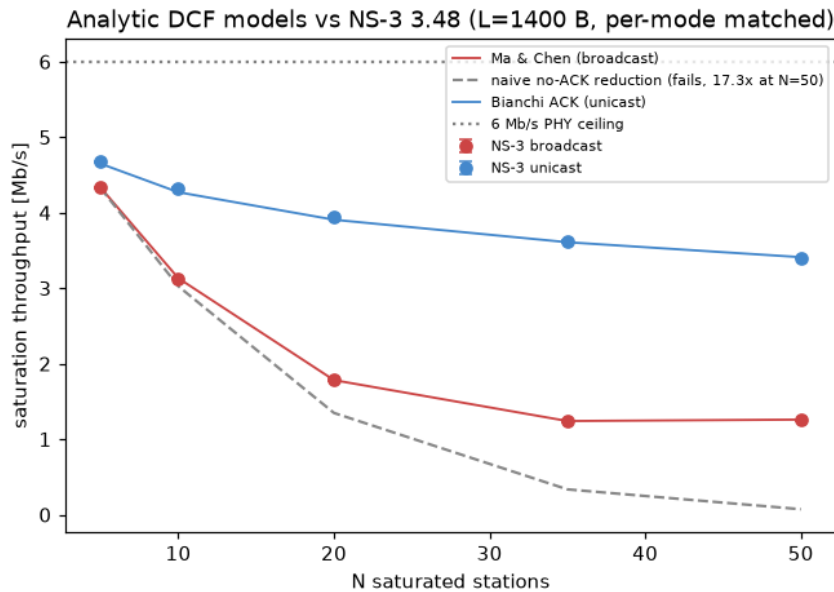


Figure 9.1: Analytic models against simulation, both modes.

so collapses it onto the naive reduction — a $16.9\times$ gap at $N = 50$. **The mechanism attribution is therefore not an inference from agreement; it is a controlled intervention.**

Remark (An expensive lesson). This project used the naive reduction — the unicast model with the acknowledgement removed and the transmission probability fixed — for several phases. It under-predicts by $16\times$ at $N = 50$. The published broadcast model’s own authors open by warning that unicast models cannot simply be reduced for broadcast analysis. *The failure was reading a model without reading its preamble.*

9.3 A non-monotonicity that is real

Saturation throughput is **not monotone decreasing** in N : it dips near $N \approx 35$ and recovers. All three implementations reproduce it, so it is a property of the freeze-process series rather than a bug — each freeze stage contributes a hump peaking at a station count set by the contention window.

This matters operationally. The capacity search breaks on the first violating N , which is valid only if utilisation is monotone. It is — because the explicit factor N in the numerator outruns the recovery — but **that is an arithmetic accident, not a structural guarantee**, and it is now verified exhaustively rather than assumed.

9.4 Hardware

Two Raspberry Pi 4 units in ad-hoc mode at 5 GHz measured broadcast link loss at $p = 2.3 \times 10^{-4}$ and airtime at 1.995 ms per frame against 1.99 ms predicted — an independent check on the timing constants underlying every airtime figure in this thesis.

Remark (Two warnings, both earned). This validates *airtime*, not contention: one transmitter means zero contention, so the broadcast contention model remains simulation-only.

And an initial 2.4 GHz sweep produced a tidy 97.45% delivery figure that was **saturation at the 802.11b 1 Mb/s broadcast basic rate**, not channel loss. It was caught by a pre-stated prediction plus a load sweep, and is kept in the record labelled as what it was. Repeatability did not protect against it — the standard deviation across windows was small. *A precisely reproducible measurement of the wrong thing looks exactly like a good measurement.*

9.5 The utilisation ceiling, and a flaw in the author’s own pre-registration

The capacity envelope applies one measured utilisation ceiling to configurations whose frames span 153–299 B, but that ceiling had only ever been measured at 288 B. A second sweep at 174 B — the compression-only baseline frame, which produces the denominators the capacity ratios divide by — tested it, with the prediction committed data-free beforehand.

	288 B	174 B
$V \geq 0.95$ crossing	2.435	2.367
combined standard error	± 0.151	
significance of the shift	0.45σ	

Statistically indistinguishable across a $1.66\times$ change in frame size. The universal ceiling is justified and no published number moves.

Remark (The prediction that should not have been made). The pre-registration also predicted a *direction* — that the crossing would drift higher — and offered a mechanism. It drifted lower, at 0.45σ . The experiment has no power to resolve the sign, so had the noise fallen the other way the author would have recorded a confirmation he had not earned, and the mechanism offered is neither supported nor refuted.

This is recorded rather than quietly dropped because it is the same defect class the rest of this thesis catalogues: **a number that looks like evidence and is not**. A directional prediction needs a power estimate, or must be stated as a band. The load-bearing prediction did carry one, which is why the run answers its question at all.

Still untested: invariance in N . Both arms are at $N = 50$ while the envelope applies the ceiling from $N = 2$ to $N = 213$. Frame-size invariance was the cheaper half of the assumption and the half that varies across the envelope's rows; the other half is stated, not measured.

Chapter 10

The Low-Rate Regime and the Exclusion Bound

► *STATUS: DRAFTED. This chapter carries the thesis’s most durable result and also its most uncomfortable self-correction. Both are stated here rather than in a limitations section.*

10.1 Why the low-rate arm is a different regime, not a slower one

The 802.11 numbers do not transfer. They differ by two to three orders of magnitude, and the binding constraint differs *in kind*: a regulatory airtime quota rather than a frame size or a latency budget. This is the third regime in the taxonomy of Theorem 4.3:

regime	what caps the batch	where
MTU-limited	frame size	—
freshness-limited	$\Lambda_i D_{\max}$	802.11
duty-cycle-limited	regulatory airtime quota	low-rate

On this arm the arrival rate and the freshness are *derived*, not configured: a frame of airtime T may repeat only every T/duty , so the sustainable rate and the staleness of the oldest record are the same interval. There is no independent deadline to trade against bytes.

10.2 The exclusion, and what part of it is arithmetic

Applying Theorem 4.6 to EU868, with $H_f = 44\text{ B}$ measured, $g_a = 64\text{ B}$ and $s_{\min} = 13\text{ B}$:

Table 10.1: The EU868 partition. **Read the tier column, not just the verdict:** it says what would have to change.

DR	payload M	$s_{\max}$	tier	durability
0, 1, 2	51 B	−57	signature	unconditional
3	115 B	7	encoding	contingent — see below
4, 5, 6	242 B	134	—	feasible

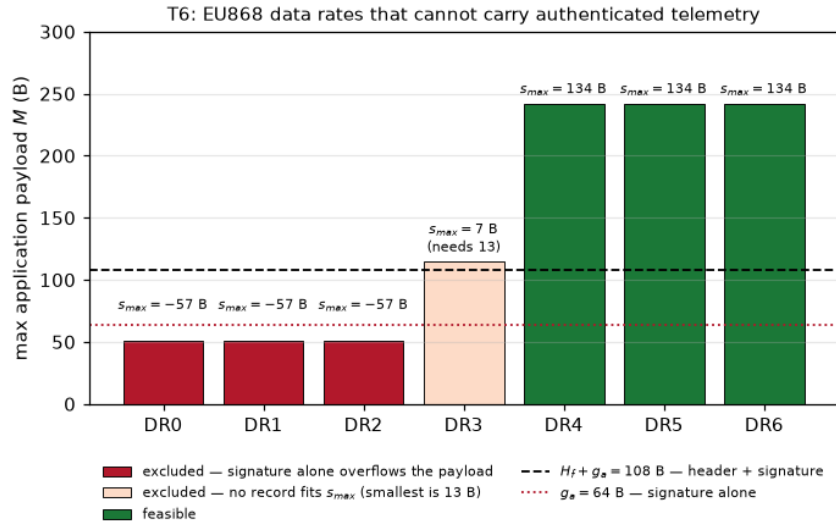


Figure 10.1: Exclusion tiers across EU868 data rates.

10.2.1 The three rates that cannot be rescued

DR0–DR2 stay excluded *with a zero-byte header and a one-byte record*. It is the signature alone that does not fit: $64 > 51$. No header design, no encoding, no batching and no chain placement reaches them. The only escape is a smaller authentication object, and the smallest standardised one at comparable security is a 48 B compressed BLS12-381 G1 point — which does fit 51 B, and leaves *three bytes* for the header and the record together.

This is arithmetic. It does not move as hardware, models or schemes improve. It is the thesis’s most durable claim and the reason the work is framed as a boundary.

10.2.2 The fourth rate, which this thesis excluded by its own framing

DR3 misses by six bytes. That margin rests on three constants, and **each of them is ours**:

1. $H_f = 44$ B, which is the *top* of a measured 38–44 B range (Table 3.2), and larger headers make exclusion more likely.
2. The payload column: published EU868 tables differ, and one source lists DR3 at

123 B where this work uses 115 B. At 123 B the residual is 15 B and **DR3 is feasible**.

3. $s_{\min} = 13$ B, the smallest record *this design* can emit.

Remark (The self-correction, stated plainly). The frame header is 29 B of CBOR *text key names*. The same seven keys encoded as small integers cost 7 B — the convention COSE and SenML already use. That single change takes H_f from 44 to 22 B, DR3’s budget from 7 B to 29 B, and **makes DR3 feasible**.

The system-model chapter had always recorded that the header was untuned. The theory chapter had always recorded that the bound depends on the header. *Nobody put the two facts side by side*. When they were, the headline moved from **four** excluded data rates to **three**.

Not even the record-field elision is required; halving the header suffices.

10.2.3 Why the smaller result is the better one

Reporting three rather than four costs a memorable number and buys three things.

It is *honest*: the four-rate claim was one reviewer’s wire-format question away from being wrong, and that reviewer would have been right. It is *durable*: what remains cannot be attacked by any framing work. And it makes the boundary **constructive** — it says not only where authentication cannot fit, but exactly what would have to change, and shows that for DR3 the answer is a header redesign rather than new cryptography.

The tier taxonomy of Theorem 4.6 always named “encoding” as the tier compression can attack. This is that tier being attacked, and moving, exactly as the taxonomy predicted.

10.3 Per-frame chaining, and why the arms differ

Moving the chain hash from per-record to per-frame is worth far more here than on 802.11, because the regional payload limit binds — so smaller records convert into *more records per frame*, which the duty cycle turns directly into sustainable rate. Measured across DR4–DR6 it gives $\approx 3.0\times$ the sustainable telemetry rate for the same regulatory budget.

The 802.11 arm keeps per-record chaining, because freshness binds there and the same change buys almost nothing in extra records — too little to pay for losing independently-transmitted per-record tamper evidence. **The two arms carry the same ledger in two framings, and that asymmetry is the point rather than an inconsistency.**

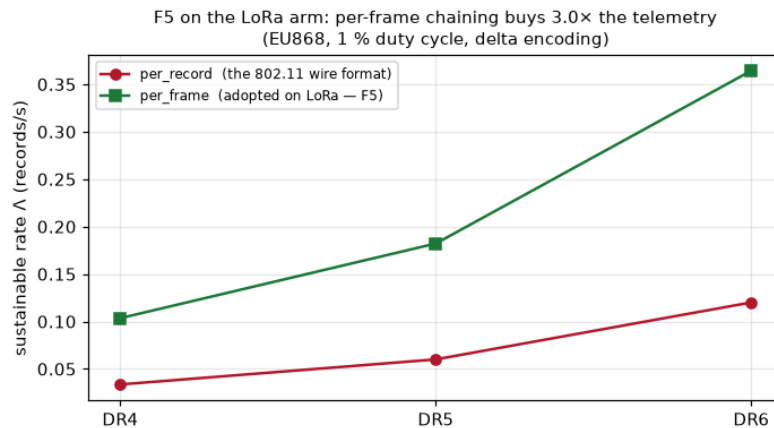


Figure 10.2: Per-frame versus per-record chaining across data rates.

10.4 Capacity, and a threshold applied to the wrong thing

Everything else in this arm is analytical and deliberately so — time on air is a specification formula and the sustainable rate is the duty cycle, both deterministic, so simulating them would be circular. Exactly one quantity resists that: *how many nodes can share the channel*, because uplinks are pure ALOHA with no carrier sense.

Simulation at DR5 gives $N_{\max} = 3$ against the $V \geq 0.95$ criterion.

Remark (Never quote this bare). The bootstrap interval is $[2, 3]$ — a knife edge — and at the certified $N = 3$, **nine of thirty seeds fail the very criterion being certified**. Under a per-realisation reading, where at least 95% of runs must individually meet V , the answer is **1**, not 3.

Both are computed and emitted. The decision to headline the mean is recorded as a decision, and it was safe to take only because the comparative result survives either reading.

An earlier value of $N_{\max} = 5$ came from a three-seed sample against a bimodal distribution and is retracted; the artifact that produced it is retained, renamed and superseded.

10.4.1 External corroboration, quoted as a curve

An external hardware-grounded model gives $N_{\max} = 4$ where this work gives 3.

Remark (Two cautions about that comparison). Their $N_{\max}$ is decided at $N = 4$ and $N = 5$, both inside the region their own fit is intercept-dominated: the polynomial predicts 1.78% loss at *zero* transmitters, roughly 40% of the entire loss budget the criterion allows. Forcing the fit through the origin would give them $N_{\max} = 5$, *widening* the gap — so quoting 4 is the conservative choice.

And the pessimism ratio is not a single number. It runs $0.91\times$ at $N = 2$, where *this work is the more optimistic model*, through $1.07\times$ at $N = 3$, to $2.1\times$ from $N = 10$ upward.

The sign changes in exactly the region where $N_{\max}$ is decided. An earlier finding in this project quoted a single ratio in the wrong direction and had to be retracted.

10.5 Scope of this chapter

This is simulation and analysis, not measurement. No radio was metered, and there is no energy column — the only measured radio power in this project is a Wi-Fi figure, and reusing it for a different transceiver would be fabrication. Airtime per record is reported instead: it is what the regulator rations, and it is computed rather than assumed.

Chapter 11

Reproducibility, Defects and Research Integrity

► *STATUS: DRAFTED. This chapter is unusual for a thesis and deliberate. It reports the errors found in this project's own results. Examiners should read it as the methodological contribution it is intended to be, not as an admission appended to a limitations section.*

11.1 Why this chapter exists

Most theses report what was found. This one also reports *what was found to be wrong, and how*. The justification is not confession but transferability: the defects catalogued below are not peculiar to this work, and several of them survived safeguards that are widely considered sufficient — version control, a green test suite, seeded runs, committed raw data.

Ten results in this project moved after they were first recorded. Only four were catchable by running more seeds. Two were *protected by passing tests*. Three were contradictions between a paper and the artifact that generated it.

11.2 The defect taxonomy

class	mechanism	instance in this work
C1	small-sample mean lands the wrong side of a threshold	four headline numbers; a capacity value $5 \rightarrow 3$
C2	an unverified constant on the measurement path	the 2.4 GHz basic-rate saturation; the frame header treated as a constant
C3	a threshold applied to a mean rather than a distribution	the capacity criterion; the utilisation crossing
C4	a configuration change that perturbs the random realisation	simulator stream displacement under mobility
C5	a claim wider than the experiment supports	two-node hardware described as validating contention

Remark (The load-bearing observation). **More seeds fixes only C1.** C2–C5 are immune to sample size. “We ran thirty seeds” is therefore not a statement of correctness, and the discipline installed after the C1 failures gave no protection at all against the C2 failures found later.

11.3 An automated staleness gate, and what it does not cover

Every model-derived artifact is re-derived from current code on every run and compared row by row; a single differing value fails the build. This makes silent staleness of *derived* data impossible.

It does not make staleness impossible. Two limits were found the hard way:

- **Scope.** Every drift guard originally parsed the paper. The specification documents were outside that scope, so corrections propagated into the paper and not into the documents for three weeks — including a table built verbatim on data the project had already purged and superseded.
- **Direction.** Retracting a finding in a register does not remove it from the prose. One retracted claim was found alive in the paper months later, one hundred lines below a sentence stating the opposite, and later again in the open-items register. **When you retract, grep the whole repository.**

11.4 Practices that earned their place

Pre-registration, committed data-free. Predictions are committed as their own commit before data exist, so the ordering is third-party checkable. This cost the project a claim once: an assertion of pre-registration was *withdrawn* rather than reconstructed, because writing an expectations file after results are known manufactures evidence.

Stating the expected value before measuring. This is what caught the 2.4 GHz basic-rate artifact, which repeatability would never have caught.

Retaining a discarded model as a labelled failure curve. The naive broadcast reduction remains in the codebase, clearly marked, so the 16× error it produces can be shown rather than described.

Keeping per-seed raw data, not summaries. This allowed validation bands to be re-derived at thirty seeds without re-simulating.

An independent second implementation. The slot-exact simulator, written before the published model was found, shares none of its assumptions and is what turns agreement into evidence.

11.5 Two defects worth stating individually

11.5.1 A test that asserted a bug

A utilisation function returned exactly zero for a single node, with a unit test asserting that a lone sender never contends. The function conflated *contention* with *utilisation*: a lone sender does not collide but still occupies airtime, and returning zero declared a single node able to carry unbounded traffic. The defect was harmless until another result began reporting single-node cases — **and it was protected by a passing test that had encoded the wrong behaviour as intended behaviour.**

11.5.2 A criterion that could not fail

The project pre-registered a success criterion requiring at least a 40% reduction in authentication bytes, and committed it to version control two days before the result existed. The ordering is genuine and verifiable.

The quantity it tests, however, reduces exactly to $1 - 1/b$. The threshold is therefore $b \geq 2$ — independent of encoding, scheme, placement, header size and signature size. The companion condition on verifiability is satisfied by construction with zero margin. **No configuration that batches at all could have failed the test.**

The pre-registration is real and worth keeping; what had to go was the implication that it constituted a risk the design might not have survived.

11.6 What a reader should take from this

Three times in this project a claim survived because two facts were each individually recorded and never placed side by side. A register that stores facts separately does not compose them. **Only re-derivation does.**

► *TO ADD: a comparison with the reproducibility literature — the simulation-credibility lineage in particular — so this chapter reads as a contribution to that conversation rather than as a project post-mortem. The companion methods paper does this; the material needs porting.*

Chapter 12

Conclusions, Limitations and Future Work

► *STATUS: DRAFTED. The future-work section is the one most likely to be examined for evidence that the candidate understands the work’s boundaries; it is written accordingly and deliberately does not oversell.*

12.1 What this thesis establishes

1. After compression, authentication — not data — is the dominant on-air cost of a tamper-evident UAV telemetry ledger. At the smallest encoding studied, the signature is larger than the record it signs.
2. Frame-level self-batching is the correct placement, and it Pareto-dominates block-level aggregation under *any* stationary loss process, not merely under the independent-loss assumption the emulator implements.
3. On IEEE 802.11 the freshness deadline binds before the frame size does, so the amplification law that motivates aggressive compression is never operative: compression pays exactly $1\times$. On low-rate links the frame size binds and the leverage is real. **The regimes are different in kind.**
4. Three of seven EU868 data rates admit no per-frame-verifiable telemetry at any encoding, batch, header or chain design, because a 64 B signature does not fit a 51 B payload. This is arithmetic and cannot move.
5. Joined to a validated channel model, the co-designed configuration sustains $1.9\text{--}3.2\times$ the neighbourhood of an inline baseline, at 58.7% fewer on-air bytes, at an operating point compliant with 3GPP TS 22.125.

6. Of the four design axes, only placement and batching genuinely interact. The four-axis co-design claim, as originally stated, was overstated.

12.2 Limitations, stated as such

12.2.1 Modelling scope

The single collision domain admits no spatial reuse and no hidden terminals; a real formation has both, one helping and one hurting. Mobility was tested and found null for collision-limited capacity, but per-frame Doppler is unmodelled — the null result means mobility does not change *collision-limited capacity*, not that mobility is harmless to a link. Loss is independent per frame; the ordering result survives correlation but absolute verifiability figures do not. The keyframe interval is fixed and unoptimised. Post-quantum sizes are projections, not implementations.

12.2.2 Validation scope

Simulation validates *throughput*. Latency, energy and loss behaviour are not validated by it, and “simulation-validated” must not be read as more. Hardware validates *airtime* on a two-node link; contention remains simulation-only. The utilisation ceiling is measured at one node count. Composed energy is a lower bound by roughly 10–14%, and the residual is uncharged prototype framing overhead — measured, stated, not corrected away.

12.2.3 The low-rate arm

Analysis and simulation only: no radio metered, no energy column, one spreading factor per run. The capacity figure is a within-one-spreading-factor bound, not a network capacity, and its confidence interval is a knife edge.

12.2.4 The boundary itself

Three of the four excluded data rates are arithmetic. The fourth was excluded by this design’s own frame header, and this thesis says so and quantifies the recovery. A reader should treat the three-rate claim as durable and the fourth as an engineering statement about *this* wire format.

12.3 Future work

12.3.1 Immediately available, and cheap

Invariance of the utilisation ceiling in N . Frame-size invariance was measured; node-count invariance was not. One additional sweep closes it, and the scenario already exists.

Pricing implicit certificates. The certificate accounting charges an explicit certificate periodically. Implicit (ECQV) certificates are the standard smaller alternative and were never modelled. The current charge is an upper bound, which cuts against this work’s own comparison — so the gap is safe but should be closed.

An integer-keyed wire profile, adopted rather than measured. Chapter 10 measures it and declines to adopt it, because the format is frozen for reproducibility. A successor should adopt it, re-freeze, and report the boundary that results — which will be a *smaller* exclusion and a better-engineered system.

12.3.2 Substantial, and worth doing

Channel-access delay in the freshness model. The freshness constraint that *sets* the batch omits the dominant delay term at high load. It is currently mitigated by reporting utilisation alongside, not by fixing the model. A DCF access-delay term would make the freshness constraint credible at high utilisation, where it currently is not.

Multi-spreading-factor capacity on the low-rate arm. Each run fixes one data rate, forfeiting the quasi-orthogonality that supplies much of a real deployment’s capacity. That answers a different question — deployment capacity rather than per-configuration feasibility — and it is the question an operator would ask next.

Contention on real hardware. Two radios validate airtime; the contention model, which carries the capacity envelope, remains simulation-only. A rig with enough nodes to reach the interesting region of the curve is the single most valuable missing measurement in this thesis.

A post-quantum arm that is implemented rather than projected. The projections show that batching cannot rescue post-quantum signature sizes, because freshness caps the batch and one signature plus header already exceeds the frame budget. That is a strong negative result and it deserves measurement rather than arithmetic.

12.3.3 Speculative, and stated as such

The exclusion bound is link-agnostic. Instantiating it on other constrained bands — and on the signature sizes the post-quantum standards actually mandate — would say which of tomorrow’s radio and cryptographic combinations are feasible before either is deployed. That is the natural generalisation, and it is the one this author would pursue.

12.4 Closing

This thesis introduces no new cryptographic primitive and makes no claim to one. Its contribution is a reproducible co-design of existing, standardised components; the quantified trade-offs that co-design entails; the boundary beyond which no choice of those components helps at all; and an account — unusually explicit for a thesis — of the errors made in establishing that boundary and the classes of defect that survived the safeguards intended to prevent them.

The most durable result here is also the smallest one that survived scrutiny: on the longest-range EU868 data rates, a 64-byte signature does not fit a 51-byte payload, and nothing in the design space changes that.

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